H

RAFFLES INSTITUTION

2013 Year 6 Preliminary Examination
Higher 1

BIOLOGY **8875/01**

Paper 1 Multiple Choice **30th September 2013**

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name and shade your Index Number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions in this paper. Answer all questions. For each question there are four possible answers **A, B, C**, and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

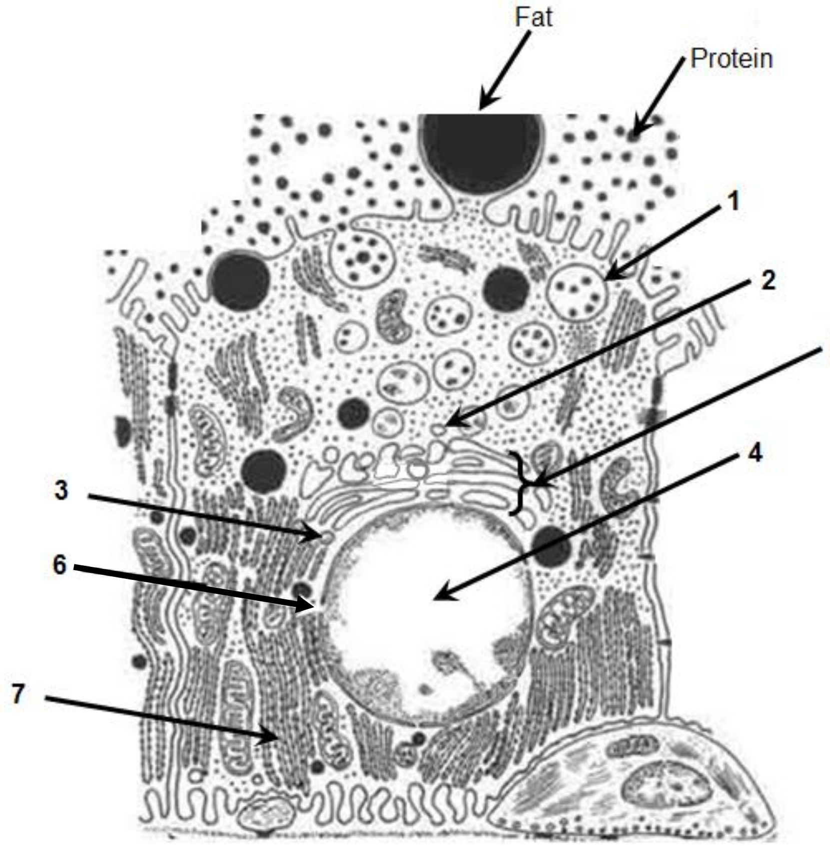
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Calculators may be used.

(Erase all mistakes completely. Do not bend or fold the OMR Answer Sheet).

This document consists of **17** printed pages.

Raffles Institution
Internal Examination

1. The diagram below shows a lactating mammary secretory cell. In this cell, milk proteins, such as casein, are transcribed and translated, and eventually secreted into the lumen of the mammary gland.



Which of the following shows the most likely sequence of locations involved in this process?

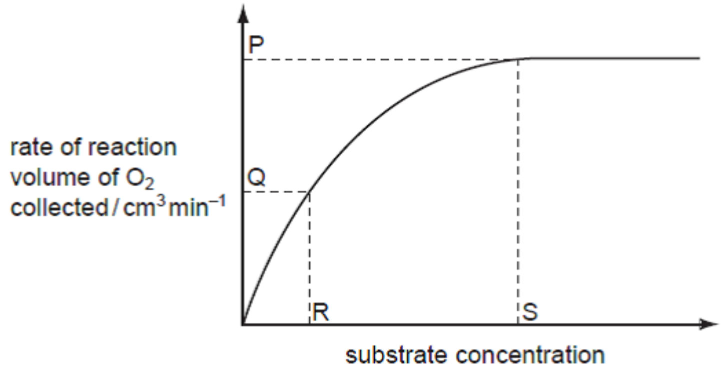
	start						finish
A	6	3	4	7	2	5	1
B	6	4	3	7	5	2	1
C	4	6	7	3	5	2	1
D	4	3	7	6	2	5	1

2. Liver tissue produces an enzyme called catalase which breaks down hydrogen peroxide into water and oxygen.

$$2H_2O_2 \rightarrow 2H_2O + O_2$$

The rate of this reaction can be determined by measuring the volume of oxygen produced in a given length of time.

Students added small cubes of fresh liver tissue to a range of hydrogen peroxide solutions and measured the volumes of oxygen produced. Their data were used to produce the graph showing how changing the concentration of hydrogen peroxide affected the rate of oxygen production.



Which statements are **correct**?

1. At **P**, the rate of reaction is limited by the concentration of enzyme.
2. At **Q**, all of the enzyme active sites are occupied by substrate molecules.
3. At **Q**, the rate of reaction is limited by the concentration of the substrate.
4. **R** represents K_m where the reaction rate = V_{max} / 2.
5. At **S**, all of the enzyme active sites are occupied by substrate molecules.

A 1, 3, 4 and 5 only

B 1, 4 and 5 only

C 2 and 3 only

D 2 and 5 only

3. Which two features contribute to the great tensile strength of cellulose?

1. glycosidic bonds linking the long chains of α-1,4-glucose molecules
2. -OH groups of the glucose molecules project outwards and form hydrogen bonds with neighbouring chains
3. strength of the glycosidic bonds between the neighbouring chains of molecules
4. successive glucose molecules are orientated at 180° to each other

A 1 and 3

B 1 and 4

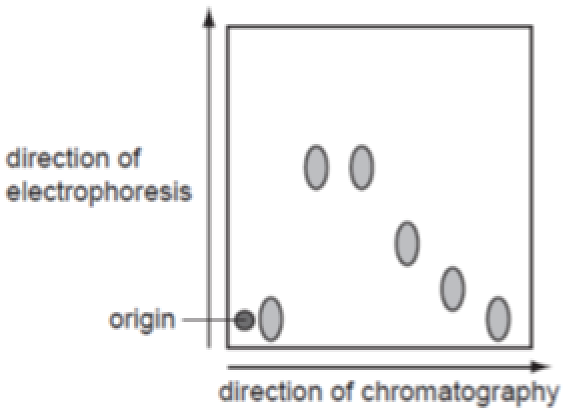
C 2 and 3

D 2 and 4

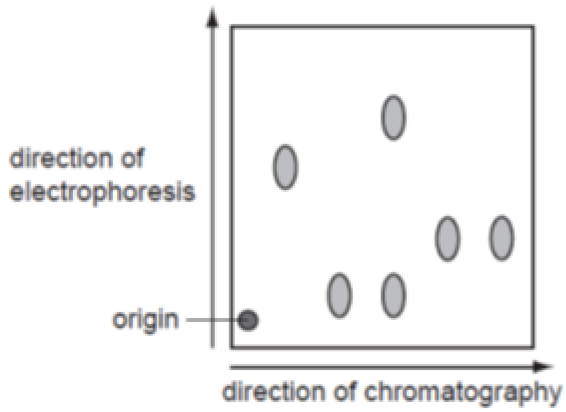
4. The diagrams below show the results of an investigation into the composition of different mixtures of amino acids. Each mixture of amino acids was separated using chromatography which separates the amino acids based on their differential solubility in a propanol and aqueous ammonia mixture. Each chromatogram was then turned through 90° and the amino acids separated again by electrophoresis. Electrophoresis separates amino acids based on their charge using an electrical field.

Which diagram shows an amino acid mixture in which the solubility of some of the amino acids is the same but the charge on those particular amino acids is different?

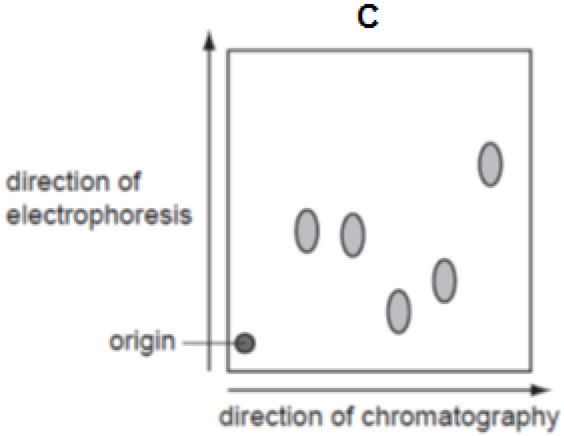
A



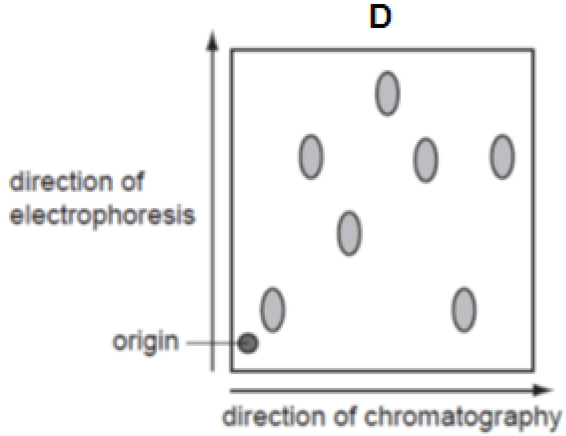
B



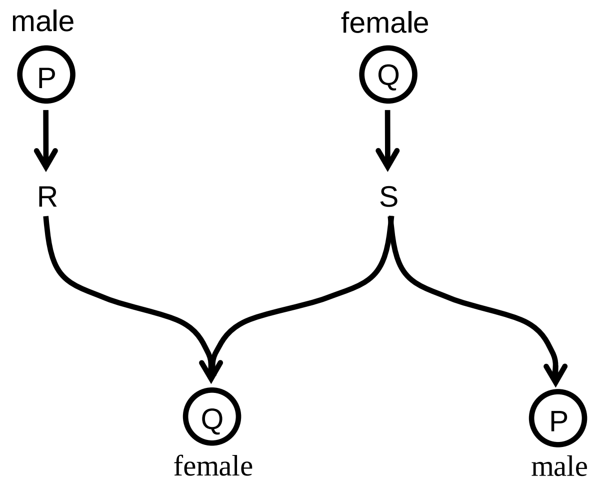
C



D



5. Sex determination in some insects such as bees and wasps is not controlled by sex chromosomes.



Using the diagram, which row in the table shows how sex is determined in these insects?

	P	Q	R	S
A	n	n	mitosis	mitosis
B	n	2n	mitosis	meiosis
C	2n	n	meiosis	meiosis
D	2n	2n	meiosis	mitosis

6. Compared to a diploid cell, how much genetic material would the nucleus of a haploid cell of the same organism contain?

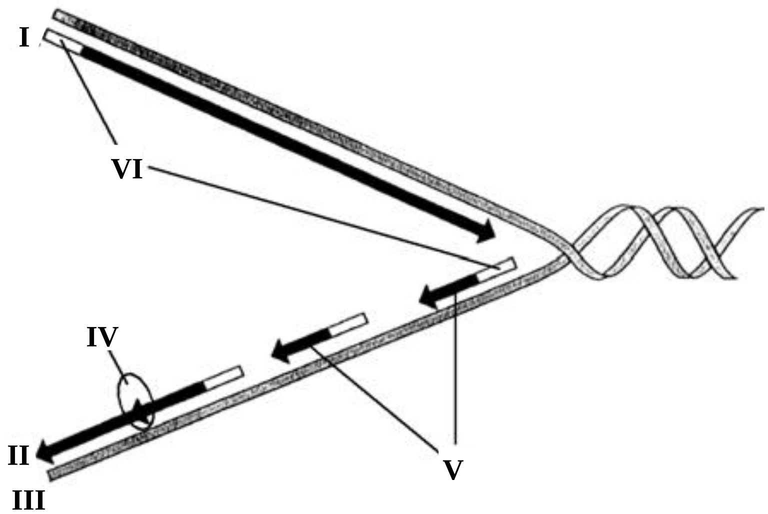
A one quarter

B double

C half

D same amount

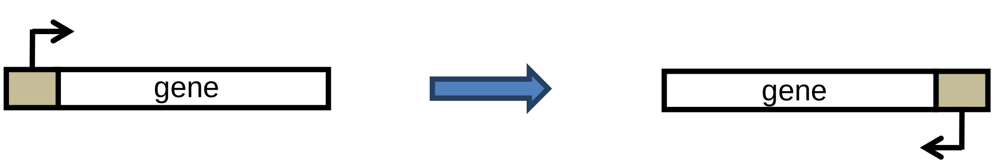
7. The diagram below shows a replication fork.



Which of the following options best describes **I**, **II**, **III**, **IV**, **V** and **VI**?

	I	II	III	IV	V	VI
A	3' end	5' end	lagging strand	DNA polymerase	Okazaki fragments	DNA primer
B	leading strand	lagging strand	5' end	DNA polymerase	Okazaki fragments	RNA primer
C	5' end	3' end	lagging strand	DNA ligase	Okazaki fragments	RNA primer
D	5' end	3' end	template strand	DNA ligase	Okazaki fragments	DNA primer

8. Scientists were able to splice out the promoter of eukaryotic gene and attach it to the end of a gene as shown below. The gene was reintroduced back to the genome of the organism.



Which of the following outcomes is **true**?

A No transcription of the gene takes place as the promoter cannot initiate transcription of the non-template strand.

B Transcription initiates but may prematurely terminate due to a premature STOP codon.

C Transcription takes place but translation cannot take place because the genetic code is no longer the original code.

D Transcription takes place but less translation takes place because the mRNA may form a duplex RNA.

9. One of the codons for the amino acid phenylalanine is UUC.

Which of the following is the **correct** anticodon of the tRNA carrying phenylalanine?

A 5’ UUC 3’

B 5’ AAG 3’

C 3’ AAG 5’

D 3’ TTC 5’

10. Below shows a partial segment of the two variant alleles of the gene coding for *EcoR1* in bacteria.

wild type	5’	Promoter	TTATACCTAATAAATGAACCTGAATTCATGCCGG 3’
	3’		AATATGGATTATTTACTTGGACTTAAGTACGGCC 5’
mutant	5’	Promoter	TTATACCTAATAAATGAACCTGAATTCATGACGG 3’
	3’		AATATGGATTATTTACTTGGACTTAAGTACTGCC 5’

Which of the following mutations is seen in this case?

A Nonsense mutation resulting in premature STOP codon.

B Silent mutation resulting in no change in amino acid coded.

C Frameshift mutation resulting in change in amino acid sequence downstream of the mutation.

D Missense mutation resulting in a different amino acid coded.

11. Duchenne muscular dystrophy is a condition characterised by progressive muscle wasting. It is caused by a recessive mutation in the DMD gene, located on the X chromosome. The DMD gene codes for a protein known as dystrophin, which, in healthy individuals, prevents damage and weakening of the muscle fibres.

Which statement explains why **not** all affected males inherit the mutation from their mother?

A Some affected males inherit the mutation from their father, who has inherited the mutation from a carrier mother.

B Some affected males inherit the normal allele of a carrier mother but synthesise dystrophin molecules that have an altered tertiary structure.

C Some males with mothers who are not carriers of the mutated allele are affected as a result of a new mutation in the DMD gene.

D The single X chromosome of some affected male becomes inactivated and no dystrophin is synthesised.

12. Achondroplastic dwarfism is an autosomal dominant disorder and red-green colour blindness is an X-linked recessive disorder.

An achondroplastic male dwarf with normal vision marries a colour blind woman of normal height. The man's father is 1.7-meters tall while both the woman's parents are of average height.

What is the probability that their son will be colour blind and of normal height?

A 0.25

B 0.5

C 0.75

D 1.0

13. *Achillea millefolium*, commonly known as yarrow is a flowering plant that was traditionally used as a herbal medicine to treat wounds and minor bleedings. It is native to temperate regions of the Northern hemisphere.

Phenotypic variation in the species has been extensively studied along an altitudinal gradient from sea level to more than 3000m elevation. It is observed that there is variation in height and biomass. Adaptation to different local environments also results in variation in physiology such as dormancy period and resistance to cold.

Which is the best method to determine if the observed phenotypic variation in the species is due to genotypic variation?

A Searching for chromosomal differences between populations at the extremes of altitude.

B Determine if viable offspring can be produced when plants from phenotypically different populations are crossed.

C Determine if hybrids between phenotypically different populations can grow at altitudes intermediate between the parent populations.

D Determine if phenotypic differences are maintained when plants from different altitudes are grown under one environmental condition.

14. A child with Down syndrome has the genotype P^1 P^2 P^3 for a polymorphism on chromosome 21 that has four different alleles — allele P^1, allele P^2, allele P^3, allele P^4. The child's mother has the genotype P^1 P^2 and the father has the genotype P^3 P^4.

In which parent did non-disjunction occur, and did this event occur in the first or second meiotic division?

A father; Meiosis I

B father; Meiosis II

C mother; Meiosis I

D mother; Meiosis II

15. Rotene and oligomycin are two metabolic poisons which affect cellular respiration. The effects of rotene and oligomycin on aerobic respiration are summarised in the table.

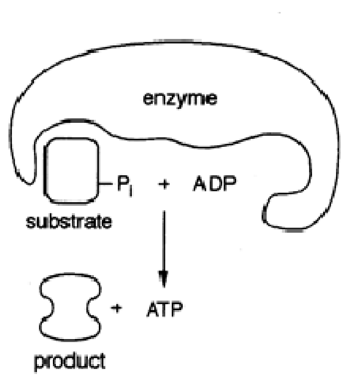
	ability to use glucose	**ability to use oxygen**	**ATP yield**
rotene	yes	no	decreases
oligomycin	yes	yes	decreases

Which of the following correctly identifies the specific functions of these two metabolic poisons?

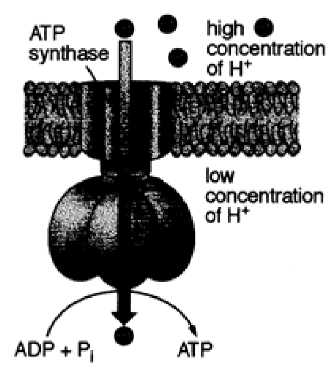
	rotene	oligomycin
A	dissipate proton gradient	inhibits ATP synthase
B	inhibits ATP synthase	electron transport inhibitor
C	electron transport inhibitor	inhibits ATP synthase
D	inhibits ATP synthase	dissipate proton gradient

16. The following 2 diagrams, **I** and **II,** are two processes involved in the complete oxidation of glucose.

I



II



Name the two processes and state the site of occurrences of these two processes.

	Name of process I	**Where process I occurs**	**Name of process II**	**Where process II occurs**
A	Substrate level phosphorylation	cytoplasm	Oxidative phosphorylation	Inner membrane of mitochondria
B	Glycolysis	mitochondria	Kreb cycle	cristae
C	Substrate level phosphorylation	Inner membrane of mitochondria	Oxidative phosphorylation	cytoplasm
D	Kreb cycle	mitochondria	glycolysis	cristae

17. The blue dye DCPIP can be converted to colourless DCPIP as shown below:

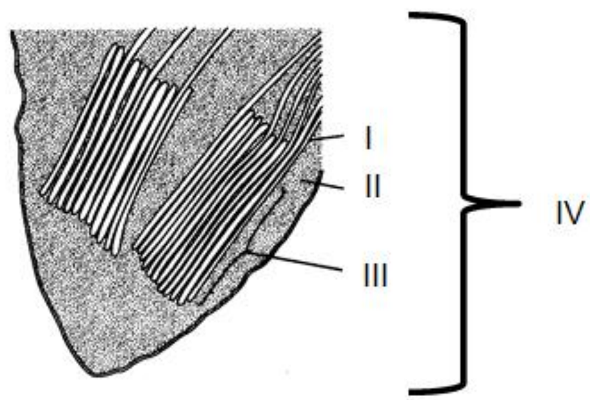
DCPIP (blue) → reduced DCPIP (colourless)

In an experiment, green chloroplast extract was first mixed with DCPIP and the extract turned blue-green. After exposure to 2 hours of light in the presence of ample carbon dioxide and water, the extract became green again.

Which of the following correctly shows the products that were formed after the experiment?

	O_2	ATP	reduced NADP	glucose
A	+	+	–	–
B	–	+	+	+
C	+	–	–	–
D	–	–	+	–

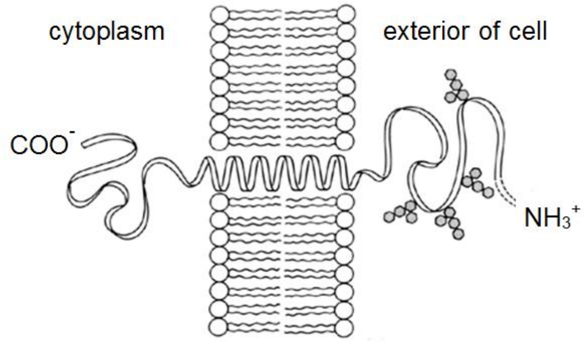
18.



Which of the following correctly identifies **I**, **II**, **III** and **IV**?

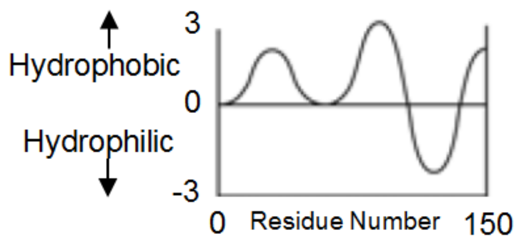
	I	**II**	**III**	**IV**
A	cristae	matrix	membrane	mitochondrion
B	thylakoid	stroma	grana	chloroplast
C	cristae	membrane	matrix	mitochondrion
D	lamella	grana	stroma	chloroplast

19. Glycophorin, an integral membrane protein, has a single transmembrane α helix.

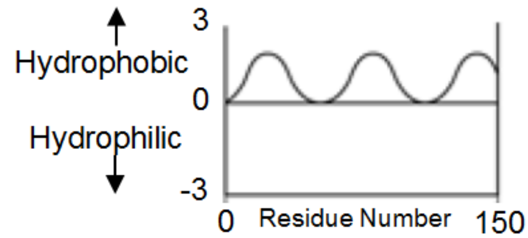


Hydropathy plots are used to estimate the relative hydrophobicity of amino acid residues. Which of the following hydropathy plot most likely represents the transmembrane nature of glycophorin?

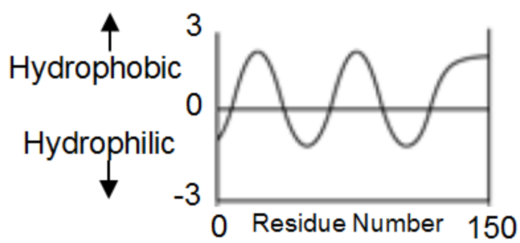
A



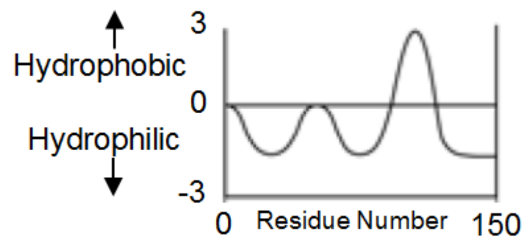
B



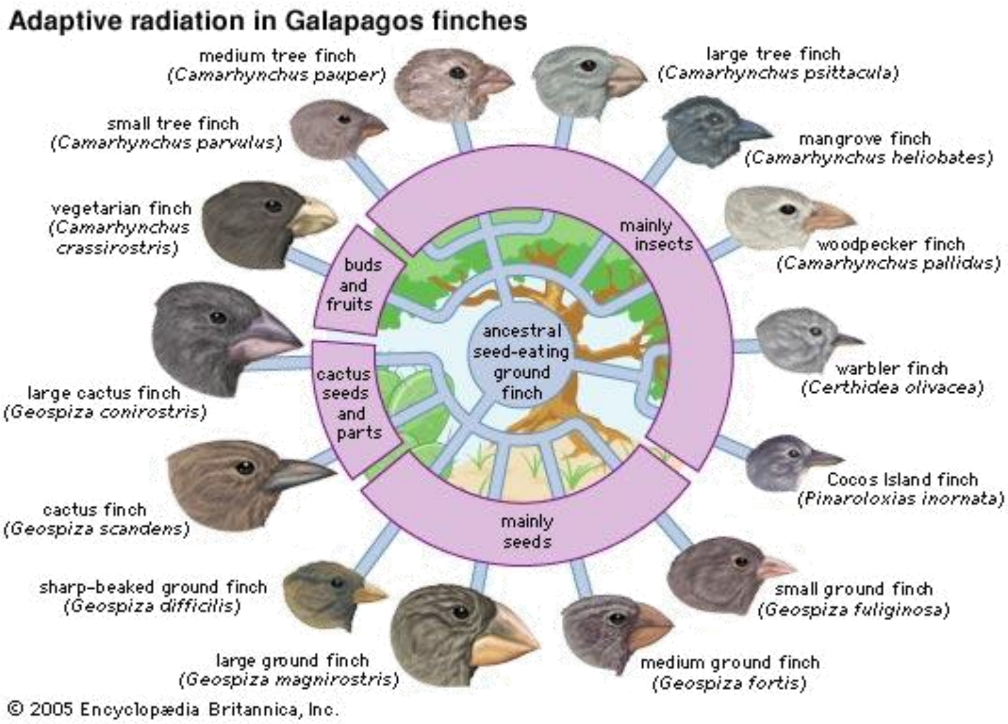
C



D



20. The Galapagos Islands are a group of volcanic islands in the eastern Pacific Ocean, about 1000km from South America. The diagram below illustrates the finches found on the islands; they resemble each other closely but differ in their feeding habits and in the shape of their beaks.



Which statement below explains how so many distinct and yet similar species arose.

A As a result of the different food sources available, the ancestral species acquired different mutations which resulted in the emergence of a variety of beak shapes.

B The finches were subjected to similar selection pressures which resulted in similar phenotypic characteristics in the distinct populations.

C The difference in food sources served as a selection pressure, allowing only finches with certain beak shapes to survive and reproduce.

D The finches migrated only to environments that they are already adapted to; finches with different phenotypes migrated to different islands.

21. Which of the following would provide the best information for distinguishing phylogenetic relationships between several species that are almost identical in anatomy?

A The fossil records

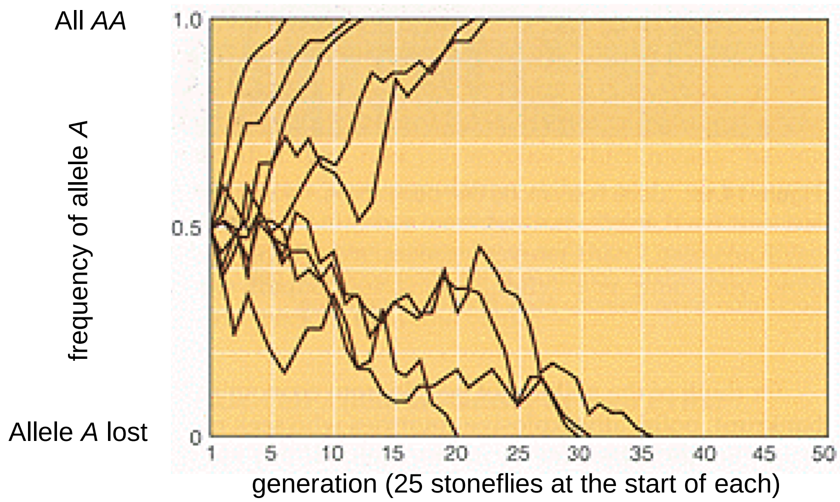
B Homologous structures

C Comparative embryology

D Molecular comparisons of DNA and amino acid sequences

22. An experiment is conducted using a small population of 25 stoneflies with the frequency of allele *A* at 50% (i.e. 1 in 2 alleles present in the population is allele *A*).

The graph below shows the changes in the frequency of allele *A* with subsequent generations. The experiment is repeated several times with the same condition and each trial is represented by a line on the graph.



Which of the following statement(s) is/are **correct**?

1. The populations were undergoing genetic drift.
2. The populations were undergoing selection.
3. Allele *A* will be lost in approximately 50% of such trials.
4. Over time, 2 separate species will emerge.

A 1 only

B 2 only

C 1 and 3 only

D 2, 3 and 4 only

23. Which of the following descriptions correctly refers to homologous structures?

	Basic structure	**Common ancestor**	**Function performed**
A	Same	No	Different
B	Different	No	Same
C	Same	Yes	Different
D	Different	Yes	Same

24. If the first three nucleotides in a six-nucleotide restriction site are **CTG**, what would the next three nucleotides most likely be?

A AGG

B GTC

C CTG

D CAG

25. Which of the following best describes the complete sequence of steps occurring during every cycle of PCR?

1. The primers hybridize to the target DNA.
2. The mixture is heated to a high temperature to denature the double stranded DNA.
3. Fresh *Taq* polymerase is added.
4. *Taq* polymerase extends the primers to make a copy of the target DNA.

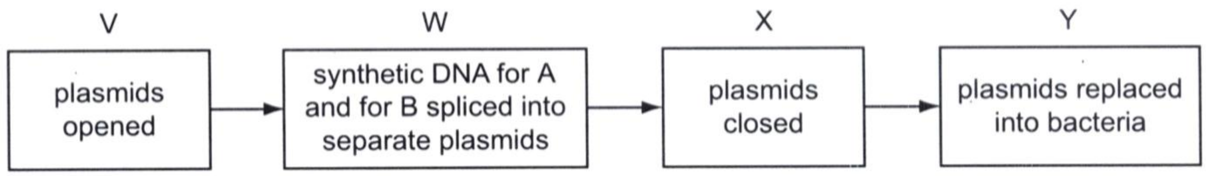
A 2, 1, 4

B 3, 2, 4

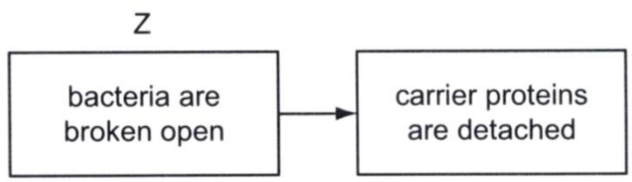
C 2, 3, 1, 4

D 2, 1, 3, 4

26. One method of making the human form of insulin uses genetically-modified *Escherichia coli*. Artificially synthesised strands of DNA which code for the insulin chains A and B and for the carrier proteins specific to chain A and chain B are spliced into bacterial plasmids. The stages are summarized in the flow chart.



The genetically-modified bacteria are grown in separate fermenters in a medium with the specific antibiotic. The two chains with their carrier proteins are synthesised inside the bacteria. Then:



Which agents are used at stages V, W, X, Y and Z?

	DNA ligase	heat shock	lysozyme	restriction enzyme
A	V	W	X and Y	Z
B	W and X	Y	Z	V
C	Y	Z and V	W	X
D	Z	X	V	W and Y

27. The human genome project has identified and mapped the genes on human chromosomes. This is allowing scientists to identify specific, faulty genes which contribute to inherited conditions.

This is useful in many ways, for example

1 carriers of faulty genes can be advised about changes in lifestyle to minimise risks.
2 carriers of faulty genes can be identified and informed of their risk status.
3 diagnostic tests can be developed to identify carriers of faulty genes.
4 drugs can be developed to block the action of problem genes.
5 embryos can be screened to avoid the birth of affected children.
6 employers can take account of the genetic predisposition of employees.

Which two uses arise **directly** from the information provided by the project?

A 1 and 2

B 2 and 5

C 3 and 4

D 5 and 6

28. Stem cells are found in many tissues that require frequent cell replacement such as the skin, the intestine or the blood.

However, within their own environments, a bone marrow cell cannot be induced to produce a skin cell and a skin cell cannot be induced to produce a bone marrow cell. Which statement explains this?

A Genes not required for a particular cell line are methylated.

B mRNA that is not required for a particular cell line is destroyed.

C Genes not required for a particular cell line are removed using restriction enzymes.

D Different stem cells have only the genes required for their particular cell line.

29. Which can best be considered to be a social implication of genetically modified (GM) crop plants?

A There is evidence that genes for herbicide resistance have spread from (GM) oilseed rape to nearby weed species although there is no evidence that these hybrids have persisted in the environment.

B In some studies, the number of beneficial insects has been reduced in areas where insect resistant GM *Bt* corn has been grown, however in other studies, there has been no effect or the number of beneficial insects has increased.

C Pressure placed by people on a multinational GM seed community has caused them to pledge that they will never develop a ‘suicide’ gene that prevents farmers from growing seed collected from last year’s crop.

D In some countries, government regulations have been put in place to prevent farmer planting GM crops close to non-GM and organic crops to prevent GM pollen from reaching the crops.

30. *Bt* gene from *Bacillus thuringiensis* may be inserted into cotton plant cells to produce *Bt* cotton plants. Insecticide use and yield in India were compared for *Bt* cotton hybrid (X*Bt*), the same hybrid X but without the *Bt* gene (X_), and another hybrid widely grown in that particular locality (Y). This process was repeated at more than 150 locations. The table below shows the results:

hybrid	**X*Bt***	**X_**	**Y**
mean number of sprays against insects that eat the cotton	0.6	3.7	3.6
mean number of sprays against sap sucking insects	3.6	3.5	3.5
yield / kg ha^{-1}	1500.0	830.0	800.0

The following are the conclusions that are drawn from the data:

1. All insect pests are killed when they consume the *Bt* crop.
2. *Bt* cotton reduces the amount of pesticide used.
3. Both yield and quality of cotton from X*Bt* crop improved.
4. *Bt* cotton increases cost effectiveness.
5. Both X_ and Y hybrids contain susceptible genes to the pest.
6. *Bt* toxin is not found in the plant sap.

Which of the conclusions stated are **correct**?

A 1, 2 and 3

B 2, 3 and 5

C 4, 5 and 6

D 2, 4 and 6

- End of Paper -